

ACID PROPERTIES

Acids

- Contain the element H.
- Produce H^{1+} ions in solution.
- Have a sour taste.
- Turn blue litmus red.

e.g. HCl - hydrochloric
 HNO_3 - nitric
 H_2SO_4 - sulfuric
 CH_3COOH - acetic

The concentration of acids (or any solution) is given by its *molarity*.

i.e. 2 M H_2SO_4 is more concentrated than 0.5 M H_2SO_4

Bases

- Any metal oxide or hydroxide.
- Produce OH^{1-} ions in solution.
- Have a soapy feel and bitter taste.
- Turn red litmus blue.

e.g. NaOH CaO
 Mg(OH)_2 NH_4OH

Note

NH_4^{1+} behaves as if it is a metal!

REACTIONS

When an acid reacts with another substance, it produces a *salt*.

Salt: non-metal part of an acid combined with a metal.

Examples

HCl - produces *chloride* salts.

e.g. MgCl_2 , NaCl , CuCl_2 , etc.

HNO_3 - produces *nitrate* salts.

e.g. $\text{Mg}(\text{NO}_3)_2$, NaNO_3 , $\text{Cu}(\text{NO}_3)_2$, etc.

H_2SO_4 - produces *sulfate* salts.

e.g. MgSO_4 , Na_2SO_4 , CuSO_4 , etc.

CH_3COOH - produces *acetate* salts.

e.g. $\text{Mg}(\text{CH}_3\text{COO})_2$, NaCH_3COO , etc.

A special reaction of acids is *neutralisation*.

Neutralisation: where a base is added to an acid (or vice versa) until the solution contains neither an acid nor a base.

In the case of a large acid spill in the environment, say on a road, emergency service workers will neutralise the acid by adding enough base to neutralise it. Then it can be safely cleaned up.